

Software Requirements Specification (SRS)

Project: Hospital Management System (HMS)

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Prepared For: ABC Multispeciality Hospital

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1. Introduction

1.1 Purpose

The Hospital Management System (HMS) is designed to digitize and streamline hospital operations, including patient registration, appointment scheduling, doctor management, billing, pharmacy, laboratory, and reporting.

1.2 Scope

The system will provide:

- Online patient registration
 - Appointment booking
 - Electronic medical records (EMR)
 - Doctor scheduling
 - Pharmacy management
 - Laboratory management
 - Billing and payments
 - Reports and analytics
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1.3 Intended Users

- Hospital Administrator
- Doctor
- Receptionist
- Laboratory Staff

- Pharmacist
 - Accountant
 - Patient
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2. Business Objective

ABC Multispeciality Hospital currently manages patient records manually, causing delays, data duplication, and reporting challenges.

The objective is to:

- Improve patient experience
 - Reduce paperwork
 - Digitize medical records
 - Automate billing
 - Improve operational efficiency
 - Generate real-time reports
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3. Functional Requirements

Module 1 – User Authentication

FR-1

Users shall log in using:

- Username
- Password

FR-2

Role-based access shall be provided.

Supported roles:

- Admin
- Doctor
- Receptionist
- Pharmacist

- Lab Technician
 - Accountant
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Module 2 – Patient Registration

The receptionist shall be able to:

- Register patients
- Update patient details
- Search patients
- View patient history

Patient information includes:

- Patient ID
 - Name
 - Age
 - Gender
 - Address
 - Contact Number
 - Blood Group
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Module 3 – Appointment Management

Patients shall be able to:

- Book appointments
- Reschedule appointments
- Cancel appointments

Receptionists shall:

- View doctor schedules
 - Allocate appointment slots
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Module 4 – Doctor Management

Doctors shall:

- View appointments
 - Add diagnosis
 - Prescribe medicines
 - Upload reports
 - Maintain patient history
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Module 5 – Laboratory Management

Lab staff shall:

- Receive test requests
- Update test status
- Upload test reports
- Notify doctors when reports are ready

Supported tests:

- Blood Test
 - Urine Test
 - X-Ray
 - MRI
 - CT Scan
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Module 6 – Pharmacy

Pharmacists shall:

- Manage medicine inventory
- Dispense medicines
- Generate invoices
- Track stock levels

System shall alert when medicine stock is low.

Module 7 – Billing

The billing module shall:

- Generate invoices
- Accept multiple payment methods
- Calculate taxes
- Apply discounts
- Print receipts

Payment methods:

- Cash
 - Credit/Debit Card
 - UPI
 - Insurance
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Module 8 – Reports

The system shall generate:

- Daily patient report
- Doctor consultation report
- Revenue report
- Pharmacy sales report
- Laboratory report
- Appointment report

Reports shall be exportable to:

- PDF
 - Excel
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4. Non-Functional Requirements

Performance

- Dashboard shall load within **3 seconds**.

- Appointment booking shall complete within **5 seconds**.
 - Reports shall generate within **10 seconds**.
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Security

- HTTPS
 - Encrypted passwords
 - Role-based authorization
 - Audit logs
 - Daily backups
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Availability

System uptime:

99.9%

Scalability

The system shall support:

- 500 doctors
 - 5,000 patients/day
 - 100 concurrent users
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Browser Support

- Chrome
 - Edge
 - Firefox
 - Safari
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5. User Roles

Role	Responsibilities
Admin	Manage users, reports, settings
Doctor	View patients, diagnosis, prescriptions
Receptionist	Register patients, manage appointments
Pharmacist	Manage inventory and medicine sales
Lab Technician	Process tests and upload reports
Accountant	Billing, payments, financial reports
Patient	Book appointments and view reports

6. Reports

Patient Report

- Total patients
- New registrations
- Patient demographics

Appointment Report

- Daily appointments
- Cancelled appointments
- Doctor schedules

Billing Report

- Total revenue
- Pending payments
- Insurance claims

Pharmacy Report

- Medicine sales
- Low stock items
- Purchase history

Laboratory Report

- Tests performed
 - Pending reports
 - Completed reports
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7. Acceptance Criteria

The project will be accepted when:

- Patients can register successfully.
 - Appointments can be booked and managed.
 - Doctors can access and update patient records.
 - Laboratory reports can be uploaded and viewed.
 - Pharmacy inventory is updated accurately.
 - Bills are generated correctly.
 - Reports can be exported to PDF and Excel.
 - All user roles have appropriate access permissions.
 - The application passes User Acceptance Testing (UAT).
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8. Assumptions

- Hospital staff are trained to use the application.
 - Internet connectivity is available.
 - Patient data is entered accurately.
 - Backup and recovery mechanisms are configured.
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9. Future Enhancements

- Mobile application (Android/iOS)
- Online video consultations
- AI-assisted disease prediction
- SMS and email reminders
- Integration with wearable health devices

- Multi-hospital support